
Degradation under Gaussian weight perturbation (γ = 0.01) vs model quality — one panel per size; small label = chinchilla
30M (0.03B)
60M (0.06B)
100M (0.1B)
300M (0.3B)
600M (0.6B)
degradation (nats)
unperturbed held-out DCLM loss
adamw
muon
30M
60M
100M
300M
600M